Pi Setup Guide

Core Objectives:

C1. Installing Raspberry Pi OS
C2. Setup Raspberry PI (Headless)
C3. Setup Raspberry PI (Screen)
C4. Connecting SSH for "headless" PI
C5. Exploring PI
C6. Testing the Pi Camera
C7. Setup NUS Wifi

Milestones (Ask Instructor / TA to check):

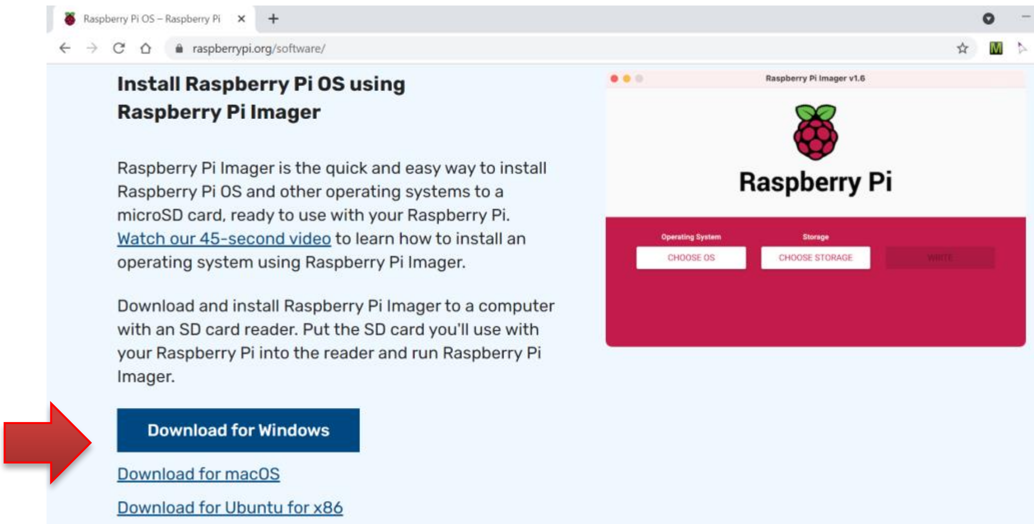
C4, C6, C7

C1. Installing Raspberry Pi OS

About Optional Pi Setup:
Section **C1 is** useful if you are starting from empty SD card. Some Raspberry Pi packages come with a pre-installed Raspbian on the SD Card, so you may not need these steps.

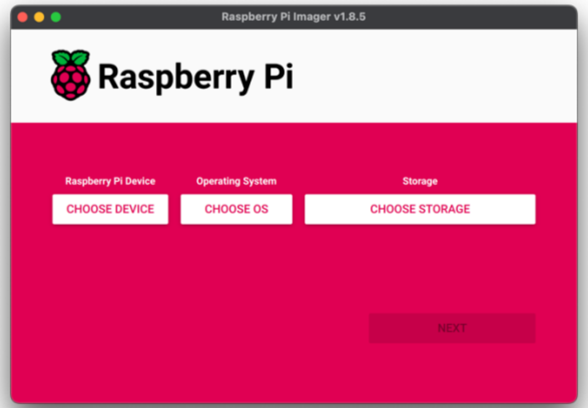
What are we doing?
Raspberry Pi, despite its cute size, is similar to all other PCs in that it requires an operating system (OS). We are going to install an OS in this section.

1. On your laptop, download and install Raspberry Pi Imager from https://www.raspberrypi.org/software/

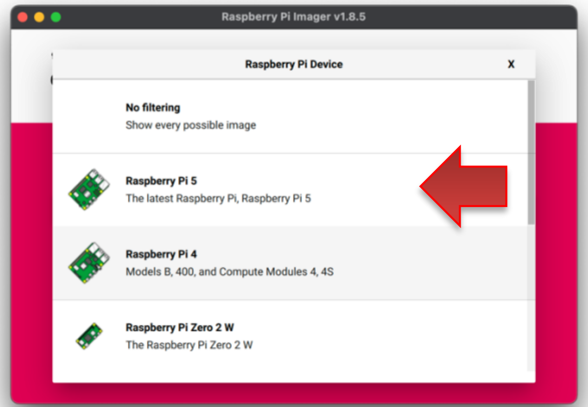


2. Slot the **16Gb SD card** into the **SD Card reader**, then plug the reader into your laptop.

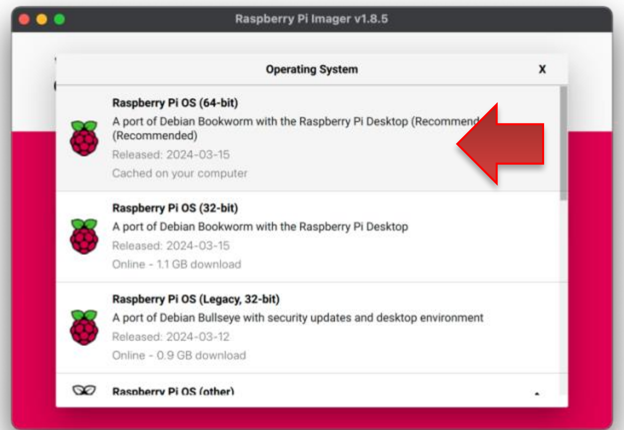
3. Start the **Raspberry Pi Imager** and you should see the following UI:



4. Click the right button "**Choose Device**" and choose the option "**Raspberry Pi 5**":

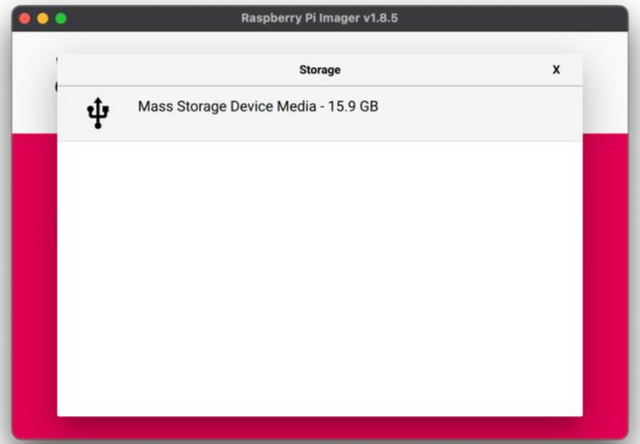


5. Click the right button "**Choose OS**" and choose the option "**Raspberry Pi OS (64-bit)**":

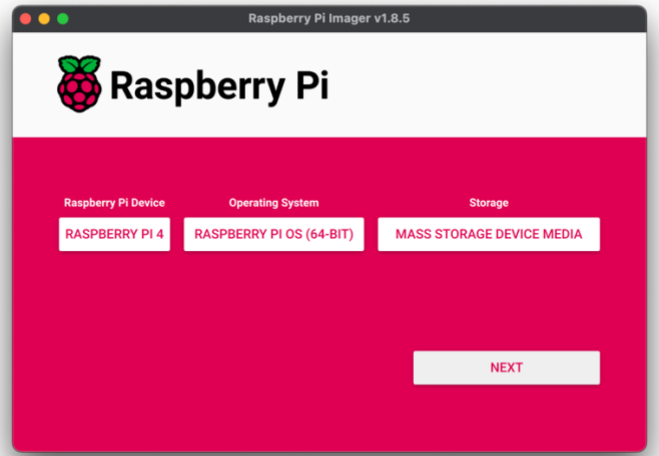


6. Plug your SD card into your SD card reader and then plug that into your computer.

7. Click the middle button “Choose Storage” to select your SD Card reader. If you are not sure, look for the drive with **~15-16GB** (your SD Card capacity) and usually appears as “**Mass Storage Device USB Device**” or “**Mass Storage Device Media**”.

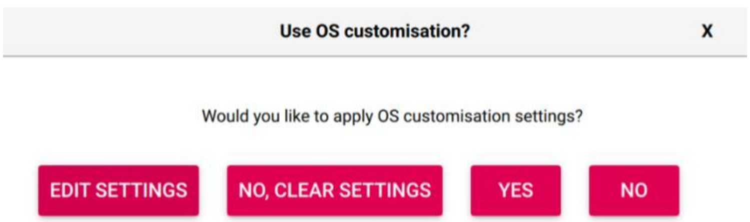


8. After the above steps, you should see something similar to this:

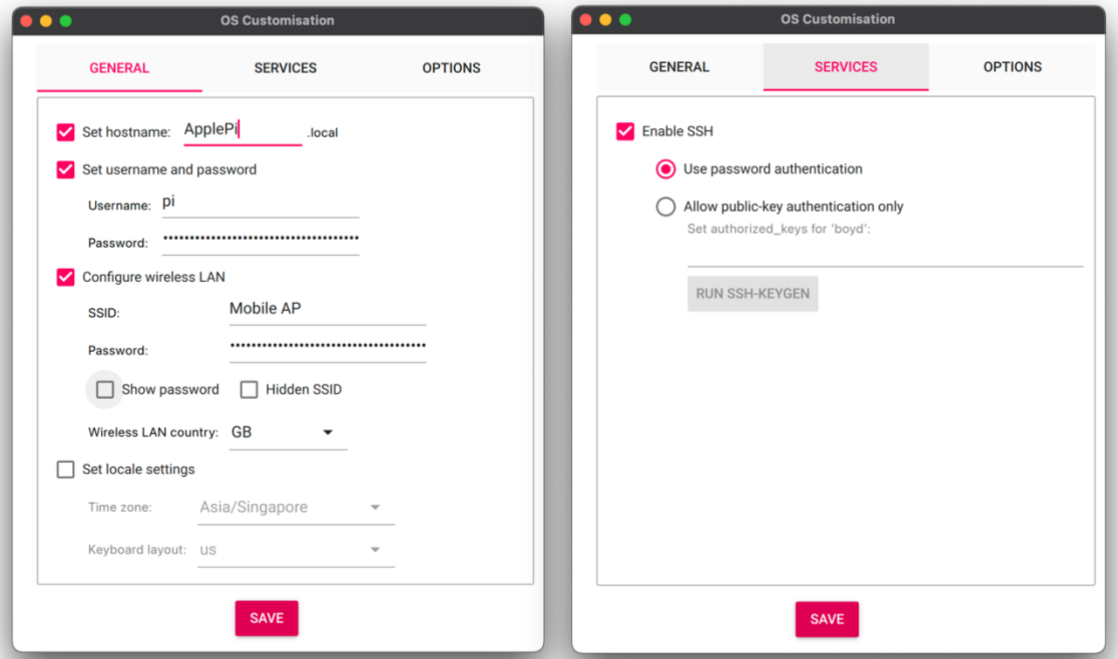


9. We are now going to set up our RPi to be headless. Click Next!

10. Now click **Edit Settings**.



You can set up your WIFI details, hostname, username and password, select the timezone and enable SSH. Don't worry if you don't have these at hand yet, we can configure these later in the process. Click **Save**.



11. Now click **Yes**. You may be asked for your password. This process may take 10-20 minutes.

C2. Setup Raspberry PI (Headless)

What are we doing?
The SD Card functions as the "hard disk" for the PI. With the Raspberry Pi OS installed, we can now start to use the Pi.

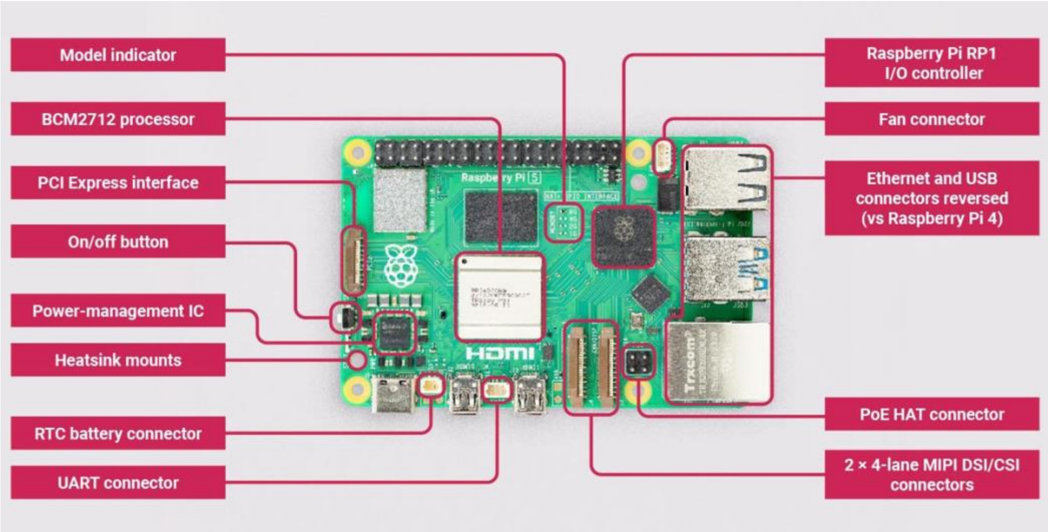


Figure 1. Raspberry PI 5 - (image taken from element14)

Steps:

1. Insert the microSD card into Raspberry Pi's microSD Card Slot ("bottom" of the Pi):

2. Let the device boot. Check your smartphone to see if it has connected to the Wi-Fi hotspot. If it has jump to C4. Otherwise, continue.

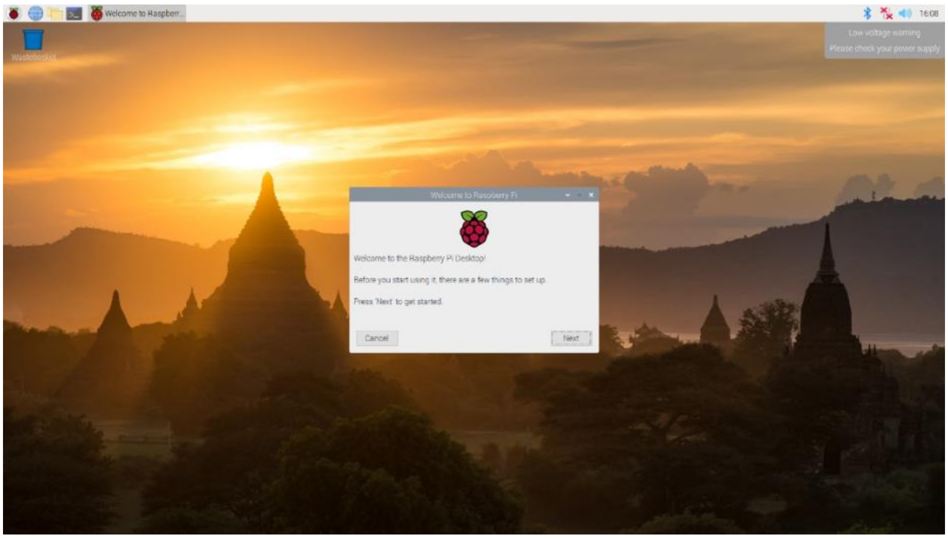
C3. Setup Raspberry PI (Screen)

3. Connect a HDMI cable to the Raspberry Pi's HDMI output then connect to the monitor. You can use your HDMI video capture card.

4. Connect a USB Keyboard + Mouse.

5. Use Micro USB power cable to connect the Pi to a power source, e.g. Wall plug or Power Bank. **Note that connecting to laptop / desktop USB port may not work well** as Pi 5 requires optimally **5 Volt at 2.5 Ampere**.

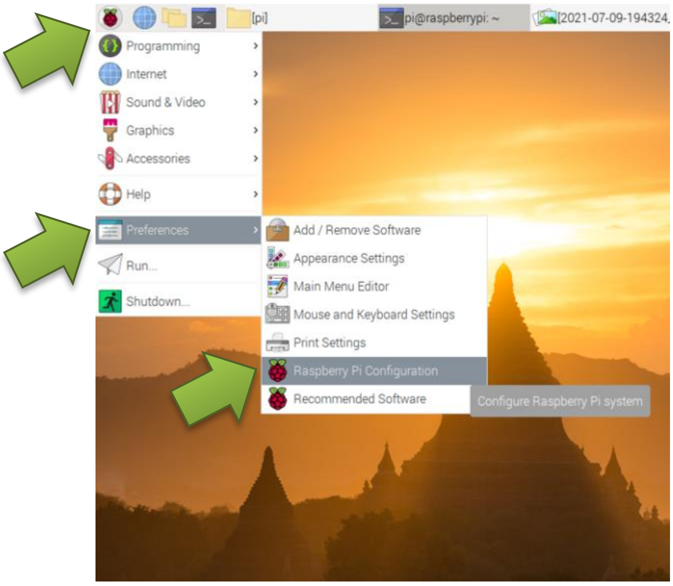
6. As soon as power is supplied, you should see PI's LEDs (Red for power, Green for SD Card reading) blinking near the Micro USB power input port. Pi should boot up with GUI Desktop similar to the following:



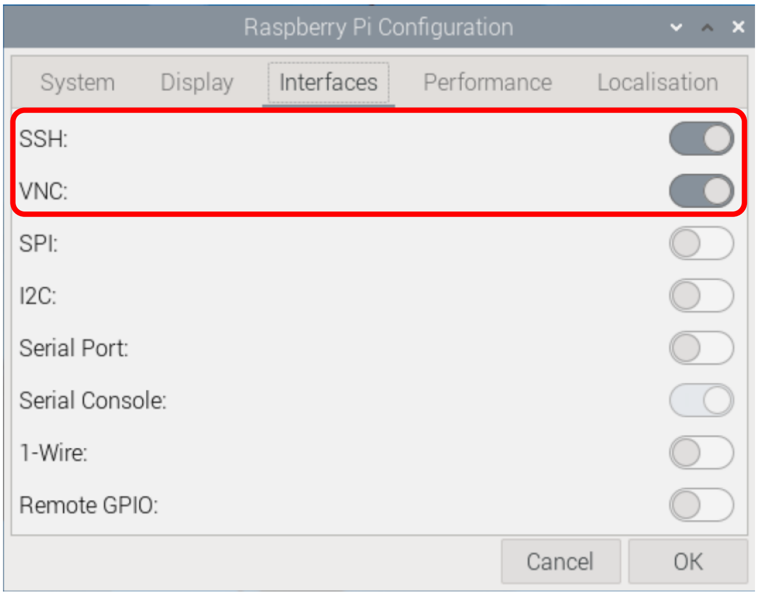
7. You will see a "Welcome to Raspberry Pi" dialog box, click **next** to continue.
   a. Select a suitable "Wifi Network" (e.g. your mobile phone wifi) and key in your wifi password. Wait for the Wifi connection to establish.

   b. We will postpone the software update for now. Click "Skip" on Update Software for now.

8. Click the upper left corner on the Raspberry icon ➔ "Preferences" ➔ "Raspberry Pi Configuration":



a. Select the "Interfaces" tab and **enable**:
   i. SSH
   ii. VNC



b. We should run the following command to enable the RPi Camera:
   i. Open the terminal and enter the following command:

```
sudo nano /boot/firmware/config.txt
```

   ii. Then add the following lines to the file:

```
start_x=1
gpu_mem=128
```

   Save the file and exit (press Ctrl+X, then press Y, and finally press Enter).

   iii. To apply the changes, restart your Raspberry Pi:

```
sudo reboot
```

iv. After restarting, you can test if the Pi Camera is working correctly by running the following command:

```
vcgencmd get_camera
```

v. If the camera is enabled correctly, you should see the following output:

```
supported=1......
```

C4. Connecting SSH for "headless" PI

Why? We are going to setup **remote connection** capability so that you can talk to the Pi through the network! **Key Idea** Essentially, if you know the IP address of the Pi, you can easily SSH into it from your laptop. This used to be quite tricky if you don't have a monitor to check the information on the Pi. Fortunately, there are now alternatives that allow you to connect without knowing Pi's IP address. **Caveat** Due to many possible configurations of your laptop / desktop, there is no single set of universal steps. We suggest two easy ways to setup.

1. Connect your Pi **and laptop** to the same wifi network.

We strongly recommend you use a VPN like Tailscale for communication, and we have created instructions for using it in the pdf: Day 1 - Tailscale and Camera Addendum.pdf

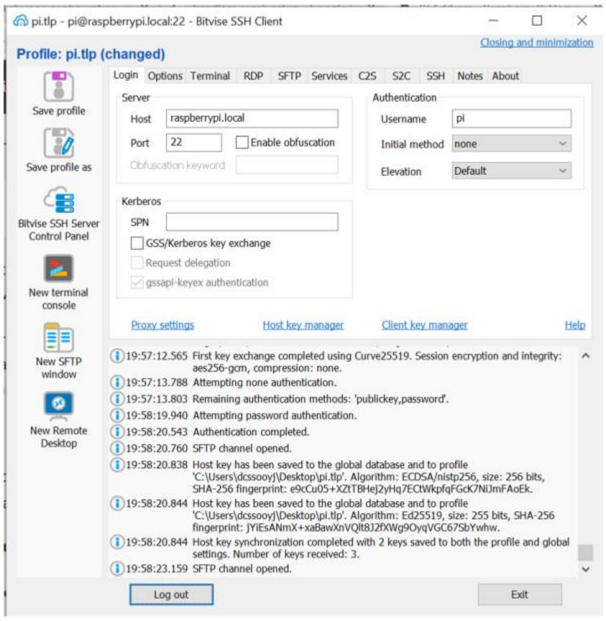
2. [Optional] Take this step just in case you need the IP address later. Check the IP address of your PI on your mobile phone. I should be something like "**192.168.X.Y**".

SSH from your laptop

On your **laptop:**

3. Install a SSH Client, e.g. bitvise SSH Client (https://www.bitvise.com/ssh-client-download) if needed.
4. There are two ways to perform the next step. (a) is the preferred, easier way.
   a. Connect to the Pi using the following information:

host: **raspberrypi.local** port: **22** username: **pi** (or the username you used during setup) password: **sws3009** (or the password you used during setup)



What is this ".local" hostname?

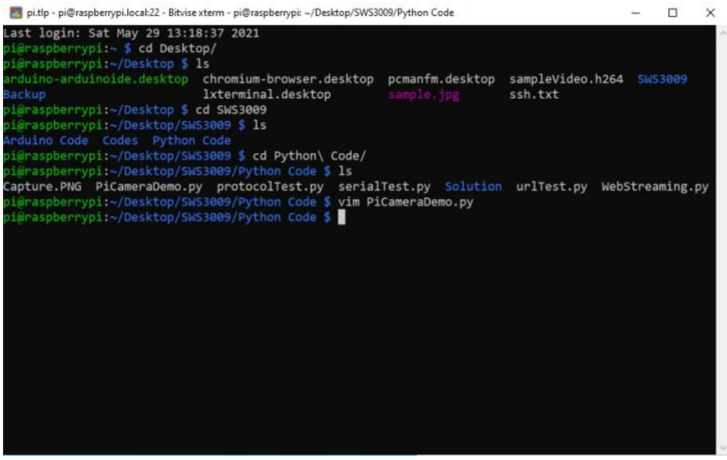
"`raspberrypi.local`" is an example of mDNS (Multicast DNS). This protocol allows a machine to "search" for a host in small network without knowing the actual IP address. Once the host received a query, it will then broadcast its IP address on the local network, which allows all other machines on the small network to pick up the IP address.

In our case, our laptop essentially asked "*who is this raspberrypi.local?*" on your wifi network. The Pi, upon receiving this query, will broadcast "*I am raspberrypi.local! My IP address is 192.168.X.Y!*". Your laptop can then use this IP address to perform the actual SSH connection.

b. **If the previous step does not work,** you need to use the Pi's IP address noted down earlier in **step 2.**

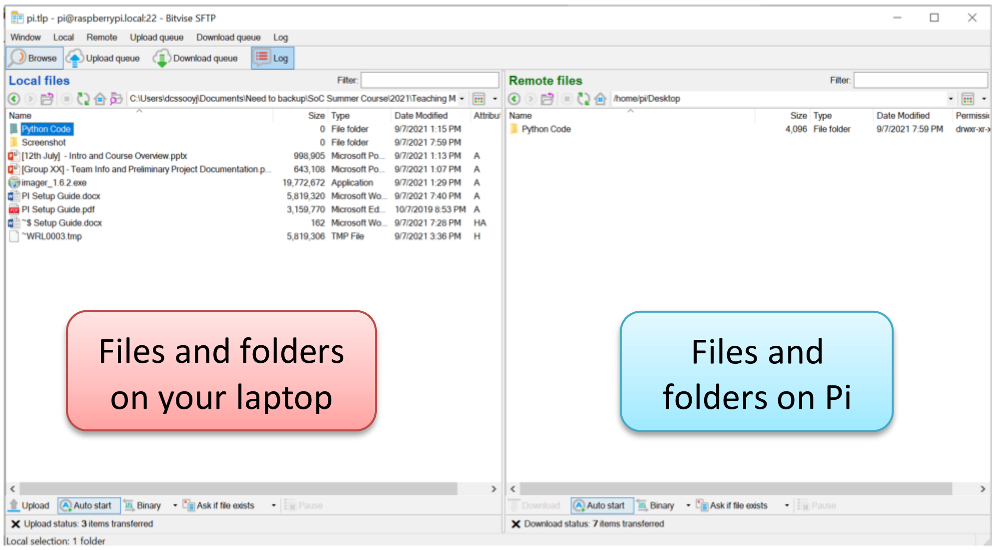
host: **(the ip address you get for the Raspberry Pi)**
port: **22**
username: **pi**
password: **sws3009** (or the password you used during setup)

5. Once connected, you can use "New terminal console" to open a terminal. This terminal is exactly the same as the terminal you can open on the Pi. Everything you type in the SSH terminal is sent over to Pi and executed there.



You now have the ability to interact with Pi via SSH on a laptop! In future, you can remove the monitor, keyboard and mouse from the Pi and use the laptop as the main way to interact.

6. In addition, you can also click “New SFTP window” to open a **file transfer interface:**



You can use drag and drop to transfer files/folders between your laptop and Pi easily. As an example, transfer the given “`Python Code/`” folder over to Pi’s desktop (`/home/pi/Desktop`).

Show the instructor / TA that you can open a terminal and have successfully transferred the “`Python Code/`” folder over to Pi.

Now, try getting VNC going. You will have to do some searching!

C5. Exploring PI

Key Idea:
Raspberry Pi is simply a very tiny computer. The major obstacle for most users is that OS used on Pi seems unfamiliar. Raspbian, the OS that we have just installed, is a **Debian**-based **Linux variant**. So, if you are familiar with other Linux/Unix based OS distributions, e.g. Ubuntu, CentOS, Fedora etc, you should be quite at home!

Raspbian come with a standard GUI desktop. Feel free to browse around in the menu options. Below are several simple tasks to familiarize you with the environment.

1. Create a **Project** folder on the desktop.
2. Create a simple `Readme.txt` file under the Project directory.
3. Find out how to write a "*hello world*" Python program and execute it.

Although it is very convenient to use a GUI environment, you are likely to use only command line input (CLI) interface for your project. Not only CLI is much lighter on the processing power / memory, it is actually equally powerful as the GUI. The only drawback is that you need to know / memorize some of useful commands.

A small set of useful commands are summarized below:

Linux Command	Functionality
`man` ***XXXX***	Get the manual (help page) on ***XXXX*** command. Useful way to learn more about a command.
`ls`	List the content of the current directory
`cd` ***YYYY***	Change directory to ***YYYY***
`cd`	Change directory to home directory
`rm` ***ZZZZ***	Delete file ***ZZZZ***
`sudo` ***AAAA***	Execute the command ***AAAA*** as the superuser. Needed for restricted command that make changes to the system. Use with care.
`apt-get`	Command to install additional packages for Raspbian. Need to use together with "`sudo`", e.g. "`sudo apt-get install vim`" which install the vim editor on Pi.
`sudo reboot now`	Reboot Pi immediately.
`sudo shutdown now`	Shutdown Pi immediately.

There are a couple of text editors installed in Raspbian by default:

1. **nano** (simple, command list on the bottom (use ctrl-<key> to activate).
2. **vi/vim** (powerful, hard to learn ☺)

C6. Testing the Pi Camera

What are we doing? For your project you are going to need to get your camera up and going! We are going to try out using the command line to take a photo and then try out

We have given some extra instructions in **Day 1 - Tailscale and Camera Addendum.pdf**

Connect your camera to the camera port on your RPi and then open a terminal window.

1. Try running the command: **rpicam-hello**
   What do you observe?

2. Now let's see how to take a photo, by running the command:
   rpicam-jpeg --output test.jpg

3. Can you figure out how to make a video using the command line?

Hint: Read the documentation!

4. However, maybe we want to take photos programmatically instead of just using the command line. Raspberry Pi provides us with a python library called PiCamera2 which lets us write a script to control the camera. We have provided you with two python scripts for you to play around with these (as well as some documentation).

   - **PiCameraDemo.py** a simple demo which takes a photo and writes some text to it!
   - **opencv_face_detect_3.py** a demo of using open cv face detection with the picamera2 library.

5. Try running these scripts (you may need to install some libraries!). Can you modify the **PiCameraDemo.py** script and change the text that is added?

Show Instructor / TA that you are able to take a picture and a video using the terminal and python scripts!

C7. Connect to the NUS WIFI

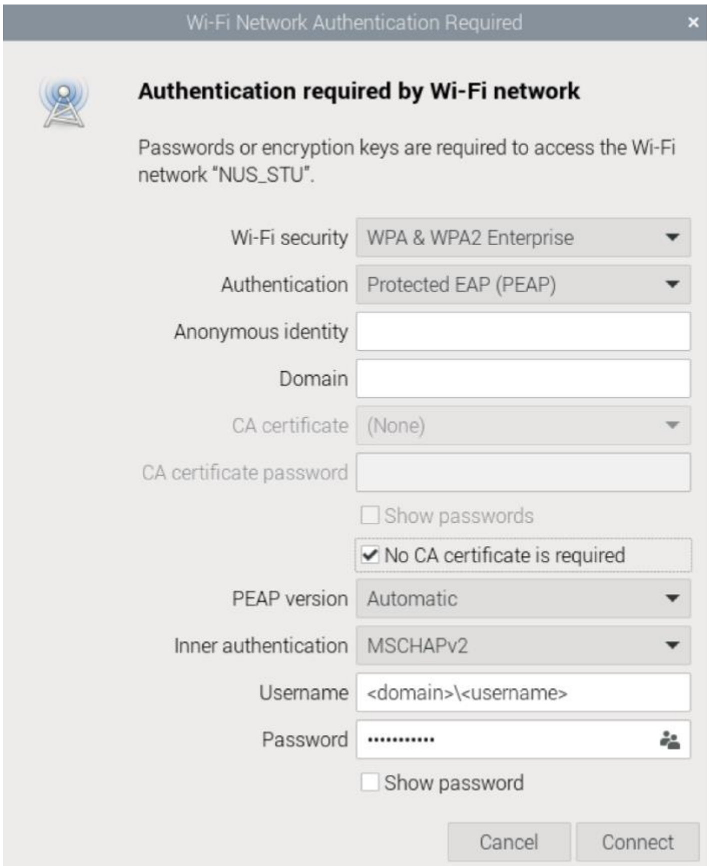
What are we doing?
Like all computing devices nowadays, Pi works much better with an internet connection. Wifi is the easiest / most common choice to setup.

Wifi networking on Pi can be quite simple for most wireless network setup. From the GUI desktop, clicking on the wifi icon on the top right corner can help you to log onto ***most*** wifi network effortlessly. In our case, the Wifi should already working after we connect to it during setup. Now try and connect to the NUS WiFi network!

If you use a GUI on your Raspberry Pi, it greatly simplifies the configuration of the WPA supplicant for you. The most important things you need to know:

• Security:	WPA2 Enterprise
• Authentication:	PEAP Select *No CA Certificate is required*
• Inner Authentication:	MSCHAPv2
• Username:	<domain>\<username> e.g. nusstu\e0985872

After playing around with your GUI, and getting your wireless connected, you can take the resulting WPA supplicant configuration to see what's needed.

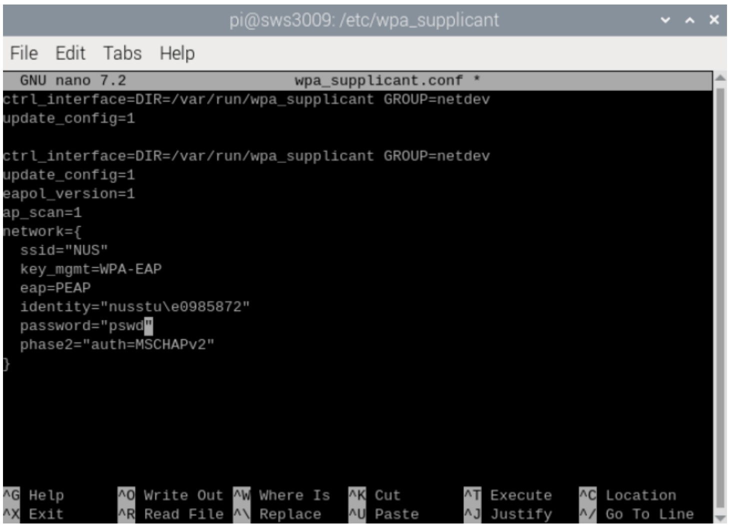


If you are using CLI version of your Raspberry Pi, you need to modify the WPA your WPA supplicant configuration file using:

```
sudo nano /etc/wpa_supplicant/wpa_supplicant.conf
```

Here's how your WPA supplicant configuration file ought to look like:

```
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1
eapol_version=1
ap_scan=1
network={
  ssid="NUS_STU"
  key_mgmt=WPA-EAP
  eap=PEAP
  identity="nusstu\a0123456"
  password="mypassword"
  phase2="auth=MSCHAPv2"
}
```



Save the file and exit (press Ctrl+X, then press Y, and finally press Enter).

With the wifi connection, you can now surf the web / perform software installation / update.

For example, get/upgrade the ***vim*** software packages for Raspbian:
- **`sudo apt-get install vim`** (if you are familiar with vim editor)
- A minimalistic of vim configuration can be found at https://github.com/mhinz/vim-galore/blob/master/static/minimal-vimrc.vim

Show Instructor / TA that you are able to surf the web (e.g. show your favourite online video?)

These **red colour** text are ***checkpoints.*** Instructor and TA will take note when you have completed these checkpoints. This is only for us to monitor the progress and render help wherever needed (i.e. please don't think of these as "tests").

When Instructor / TA join your study group, show them these checkpoints. You can attempt later checkpoints without waiting for the instructor / TA.